

Exp = 12.1

Write a program to define some constant variable in preprocessor

→

```
#include <stdio.h>
```

```
#define PI 3.14159
```

```
#define MAX_STUDENTS 50
```

```
#define GREETING "Welcome to C programming!"
```

```
int main() {
```

```
    printf("%s\n", GREETING);
```

```
    printf("The value of PI is %.2f\n", PI);
```

```
    printf("Maximum number of students allowed: %d\n", MAX_STUDENTS);
```

```
    return 0;
```

```
}
```

FileEditSelection...<=>c programming

exp1.c exp_12.c Xexp2.c exp_12.cexp1.c exp_13.cexp2.c exp_5.c

exp_12.c > C exp1.c > main()

```
1  #include <stdio.h>
2
3  #define PI 3.14159
4  #define MAX_STUDENTS 50
5  #define GREETING "Welcome to C Programming!"
6
7  int main() {
8      printf("%s\n", GREETING);
9      printf("The value of PI is: %.2f\n", PI);
10     printf("Maximum number of students allowed: %d\n", MAX_STUDENTS);
11
12     return 0;
13 }
```

PROBLEMSOUTPUTDEBUG CONSOLETERMINALPORTS

C/C++ Compile Run +

PS C:\Users\sanch\OneDrive\Desktop\c programming> cd 'c:\Users\sanch\OneDrive\Desktop\c programming\exp_12.c\output'
PS C:\Users\sanch\OneDrive\Desktop\c programming\exp_12.c\output> & .\exp1.exe
Welcome to C Programming!
The value of PI is: 3.14
Maximum number of students allowed: 50
PS C:\Users\sanch\OneDrive\Desktop\c programming\exp_12.c\output>

Ln 13, Col 2Spaces: 4UTF-8CRLF{ } CSigned outWin32

Compilation successful.

Exp 12.2

→ Write a program to define functions in
directives

→

```
#include <stdio.h>
```

```
#define SQUARE(x) ((x) * (x))
```

```
int main () {
```

```
    int a = 6;
```

```
    int b = 3;
```

```
    int result 1, result 2
```

```
    printf (" --- Function-like Macro Demonstration --- \n");
```

```
    result1 = SQUARE(a);
```

```
    printf (" SQUARE(%d) = %d \n", a, result1);
```

```
    result2 = SQUARE(a + b);
```

```
    printf (" SQUARE(%d + %d) = %d \n", a, b, result2);
```

```
    return 0;
```

```
}
```

